Assume/Guarantee Contract of `PseudoRate`

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1 Variable Declaration

Maps : `variables` in Code $\mapsto$ variables in Formal Specification

Alphabet : pu, Yp′, Yn′, r, r′, h1, h1′, clock, clock′

Inputs :

- `pu` $\mapsto$ pu : $\mathbb{N} \rightharpoonup \mathbb{R}$

Outputs :

- `Yp` $\mapsto$ Yp : $\mathbb{N} \rightharpoonup \mathbb{R}$
- `Yn` $\mapsto$ Yn : $\mathbb{N} \rightharpoonup \mathbb{R}$

In-Outputs : None

States :

- `r` $\mapsto$ r : $\mathbb{N} \rightharpoonup \mathbb{R}$

Parameters :

- `h1` $\mapsto$ h1 : $\mathbb{N} \rightharpoonup \mathbb{R}$

2 AG Contract

Assumptions :

$$\begin{aligned} Assumptions \triangleq\ & \mathsf{len}(\mathsf{pu}) \geq 3 \\ & \wedge\ \mathsf{len}(\mathsf{r}) = \mathsf{len}(\mathsf{h1}) = 3 \end{aligned}$$

Guarantees :

$$Guarantees \triangleq Check_1 \wedge Check_2$$

$$Check_1 \triangleq \mathsf{len}(\mathsf{Yp}') = \mathsf{len}(\mathsf{Yn}') = \mathsf{len}(\mathsf{r}') = \mathsf{len}(\mathsf{h1}') = 3 \wedge \mathsf{h1}' = \mathsf{h1} \wedge \mathsf{clock}' = \mathsf{clock}$$

$$Check_2 \triangleq \forall 0 \leq i < 3 \cdot \begin{pmatrix} \mathsf{if}\ \mathsf{pu}[i] - \mathsf{r}[i] > \mathsf{h1}[i]\ \mathsf{then} \\ \quad \mathsf{Yp}'[i] = \mathtt{0x0} \\ \quad \wedge\ \mathsf{Yn}'[i] = \mathtt{0xF} \\ \quad \wedge\ \mathsf{r}'[i] = 0.9231 \times \mathsf{r}[i] + 0.07688 \\ \mathsf{elif}\ \mathsf{pu}[i] - \mathsf{r}[i] < -\mathsf{h1}[i]\ \mathsf{then} \\ \quad \mathsf{Yp}'[i] = \mathtt{0xF} \\ \quad \wedge\ \mathsf{Yn}'[i] = \mathtt{0x0} \\ \quad \wedge\ \mathsf{r}'[i] = 0.9231 \times \mathsf{r}[i] - 0.07688 \\ \mathsf{else} \\ \quad \mathsf{Yp}'[i] = \mathtt{0x0} \\ \quad \wedge\ \mathsf{Yn}'[i] = \mathtt{0x0} \\ \quad \wedge\ \mathsf{r}'[i] = 0.9231 \times \mathsf{r}[i] \\ \mathsf{endif} \end{pmatrix}$$